Hazard Analysis
TeLaC

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Table 1: Revision History

Date	**Developer(s)**	**Change**
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

Contents

1 Introduction **1**

2 Scope and Purpose of Hazard Analysis **1**

3 System Boundaries and Components **1**
- 3.1 ML Transcription Model . . . 1
- 3.2 Database . . . 1
- 3.3 Account System . . . 2
- 3.4 Web Platform . . . 2
- 3.5 Report Generation . . . 2
- 3.6 Video Recording . . . 2

4 Critical Assumptions **2**

5 Failure Mode and Effect Analysis **3**

6 Safety and Security Requirements **7**
- 6.1 Security Requirements . . . 7
- 6.2 Safety Requirements . . . 7

7 Roadmap **8**

8 Appendix — Generative AI Use Disclosure **9**
- 8.1 AI Tools Used . . . 9
- 8.2 How and Where Used . . . 9
- 8.3 Verification of AI-Generated Content . . . 10

9 Appendix — Reflection **11**

1 Introduction

TeLaC is a telehealth platform for assessing the speech of bilingual children at home. Patients complete speech assessments online, and speech clinicians use the platform to review the recordings, verify the results, and manage their patients' records and analytics. TeLaC streamlines the clinician's workflow by combining speech annotation and assessment analysis in one system. Annotation is supported by automatically generated transcripts that clinicians can review and correct, and analysis is supported by organized, visualized results that point out behaviors needing the clinician's attention.

Because TeLaC handles sensitive health information and supports clinical decisions about children, failures in the system could have serious consequences. A hazard is defined as any condition or state of the system that, under certain circumstances, could lead to harm to patients, such as an incorrect clinical conclusion or the disclosure of sensitive information. This document identifies the hazards in TeLaC, analyzes how they could arise, and describes how the system's design and requirements reduce them.

2 Scope and Purpose of Hazard Analysis

This document examines the hazards in each component of TeLaC outlined below. For each hazard, the analysis determines the possible failures, along with their effects, causes, and recommended actions. It covers both technical hazards and user interaction hazards, because clinicians rely on the system's outputs when assessing a child's language ability, and the information it stores is confidential. If these hazards are not addressed, patients and clinicians could suffer losses such as the exposure of private health information, incorrect clinical conclusions that affect a child's care, the permanent loss of assessment data, and a loss of confidence in the platform.

3 System Boundaries and Components

[Dividing the system into components will help you brainstorm the hazards. You shouldn't do a full design of the components, just get a feel for the major ones. For projects that involve hardware, the components will typically include each individual piece of hardware. If your software will have a database, or an important library, these are also potential components. —SS]

The system can be separated into six major components, some of which work together to form the complete system, as described below.

3.1 ML Transcription Model

The ML (Machine Learning) Transcription Model is an external speech recognition service (OpenAI's Whisper) that MERLS uses to automatically convert the audio of recorded assessments into timestamped text. The system sends audio segments to the service and receives the transcribed text back along with segment timings and reliability values.

3.2 Database

The database stores all of the data that the system relies on, including patient and clinician profiles, assessment recordings, transcripts, assessment annotations and notes, and generated reports. This

component also stores video recordings and sensitive health records, making it one of the most sensitive parts of the system.

3.3 Account System

The account system handles authentication and role-based access control. This component verifies each user's identity during sign-up and login. It assigns each account a role, such as clinician or patient, which determines the parts of the interface and functionality that account role is able to access.

3.4 Web Platform

The web platform is responsible for hosting the patient-facing interface and the clinician interface together. The clinician interface includes a clinician dashboard, where clinicians are able to view their patients and assessment history. It also includes the annotation interface where assessment videos and their transcriptions are displayed and ready for annotation. The web platform maps user actions to the other system components and displays their results.

3.5 Report Generation

This component compiles the results of an assessment into a report that clinicians can review and use when evaluating any patient. It compiles the automatically generated results and flags, along with transcription edits, annotations, and notes made by the clinician, so that the final report accurately reflects the clinician's evaluation. Once generated, reports are saved to the database and shown on the clinician's dashboard.

3.6 Video Recording

The Video Recording component captures the video and audio of a live assessment session between a clinician and a patient, and uploads the recording to the system once the session ends.

4 Critical Assumptions

[These assumptions that are made about the software or system. You should minimize the number of assumptions that remove potential hazards. For instance, you could assume a part will never fail, but it is generally better to include this potential failure mode. —SS]

- Assume that the clinicians using the system are licensed speech-language pathologists.
  - This assumption is made since the system is designed to support clinical assessment. If the clinician is not licensed, then they should not be assessing or diagnosing a child's speech and language abilities.
- Assume that the clinicians have access to an approved device with stable internet.
  - This assumption is made since the system is designed to be used on a desktop or laptop with a web browser. If the clinician doesn't have access to such a device or stable internet connection, they would not be able to use the system.

- Assume that clinicians review the system's flags and transcripts before relying on them, so that they make the final judgement.
  - This assumption is made since the system is designed to assist clinicians, not replace them. If clinicians rely on automatically generated results without reviewing them, then errors in transcription or flagging may lead to incorrect conclusions about a child.
- Assume that patients and caregivers have consented to the assessment being recorded and stored.
  - This assumption is made since the system records and stores video of children. If consent has not been given, then the recording should not be collected or stored by the system, nor should any assessment be conducted via the system.
- Assume that the assessment contents are provided by a professional research team and are clinically correct.
  - This assumption is made since the system is designed to support real, professional clinical assessments and the team isn't responsible for designing the assessments themselves. If the assessment content is incorrect, then the results shown to clinicians may be misleading even if the system works as intended.
- Assume that assessments are conducted in languages supported by the transcription model.
  - This assumption is made since the system relies on an existing speech recognition model. If an assessment is conducted in an unsupported language, then the system will not be able to generate a meaningful transcript.
- Assume that the clinician has instructed a patient with speech or language difficulties to use this system, and that the patient is able to use the system without any assistance.
  - This assumption is made since the system is designed to be used by patients with speech or language difficulties. If the patient is not able to use the system without significant assistance from another person, then the assessment may not be completed accurately.

5 Failure Mode and Effect Analysis

[Include your FMEA table here. This is the most important part of this document. —SS] [The safety requirements in the table do not have to have the prefix SR. The most important thing is to show traceability to your SRS. You might trace to requirements you have already written, or you might need to add new requirements. —SS] [If no safety requirement can be devised, other mitigation strategies can be entered in the table, including strategies involving providing additional documentation, and/or test cases. —SS]

The following FMEA table is a breakdown of the hazards that could occur within the system, along with recommended actions to mitigate them.

Table 3: Failure Mode and Effect Analysis

Component	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
ML Transcription Model	1. Model omits speech or invents text that was not said 2. Audio is sent to the external API without adequate protection	1. Missed or hallucinated utterances falsify the child's speech assessment, which can lead to incorrect conclusions about their abilities. 2. Private recordings of children's assessments are exposed, which is a breach of sensitive data and a potential violation of privacy regulations such as HIPAA.	1. Low-volume audio, silence or noise misread as speech, or model hallucination. 2. Unencrypted transmission, or data leaks, since OpenAI may store submitted data for up to 30 days (**?**).	1. Flag segments the model is less certain about and prompt the clinician to review them against the video playback. 2. Encrypt transmissions, and apply for Zero Data Retention to prevent OpenAI from keeping the recordings (**?**).	1. FR13 2. NFR7	1. HA-MLTM1 2. HA-MLTM2
Database	1. Patient data is accessed by an unauthorized user 2. Saved data is lost or has errors 3. Data gets linked to the wrong patient or assessment	1. Recordings and health records of children are exposed, which is a breach of sensitive data and a potential violation of privacy regulations like HIPAA. 2. Recordings, assessment, or annotation data is lost and cannot be recreated, so the clinician must redo the assessment. 3. The clinician works on another patient's assessment by accident, which can lead to an inaccurate assessment and report about a child.	1. Missing server-side authorization checks, injection attacks, unencrypted storage, or stolen credentials. 2. Storage failure, interrupted or partial writes, concurrent edits overwriting each other, or accidental deletion. 3. Incorrect record identifiers and errors when saving an uploaded recording.	1. Encrypt stored data, verify on the server that the clinician is assigned to the patient before returning any record, use parameterized queries, and log every access. 2. Ensure that data is backed up regularly, with confirmation of each save to the clinician, while keeping the original auto-generated version of edited content. 3. Enforce links between patients, assessments, and recordings in the database, and display the patient's name and assessment date on every assessment view.	1. HA-SER3 2. NFR9 3. FR19	1. HA-DB1 2. HA-DB2 3. HA-DB3

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Table 3 Continued from previous page

Component	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Report Generation	1. Report does not accurately show the clinician's latest changes 2. Unverified automated content appears on the final report 3. Downloaded report is exposed to unauthorized people	1. The report misrepresents the clinician's evaluation, and incorrect results may be shared with families, schools, or other professionals. 2. A transcription or flagging error is presented as the clinician's judgement, which can lead to an incorrect diagnosis. 3. A child's assessment results and health information leave the system's protection and are shown to unauthorized people.	1. The report is generated from outdated data because edits were not saved before generation, or the assessment changed after the report was generated. 2. The clinician generates the report before reviewing all automated content, or the system lets the report be finalized without the required checks. 3. The file is sent to the wrong recipient, stored on a shared or lost device, or downloaded by a user who is not the assigned clinician.	1. Save all edits before generating the report, stamp each report with its generation date and time, mark it as out of date if the assessment changes, and show a preview before download. 2. Label unverified content in the report, and prevent the report from being finalized until the clinician has reviewed all automated content. 3. Allow only the assigned clinician to download reports, record every download, and mark each report as confidential.	1. FR7 2. HA-SAR2 3. NFR5	1. HA-RG1 2. HA-RG2 3. HA-RG3
Video Recording	1. Poor audio or video quality 2. Recording is saved but corrupted or unplayable	1. Transcription and face or behavior detection are unreliable, so the assessment may be wrongly critiqued and analyzed. 2. The clinician cannot review the assessment, and the data cannot be recreated.	1. Low-quality microphone, background noise, poor lighting, low-quality camera. 2. Interrupted upload, format or encoding error, storage failure.	1. Provide a setup tutorial and run a quality check before conducting the assessment. 2. Keep a temporary copy of the recording in the patient's browser until the system confirms the upload succeeded, verify that the recording is complete and playable, and back up the recording once it is verified.	1. NFR9 2. NFR10	1. HA-VR1 2. HA-VR2

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Table 3 Continued from previous page

Component	Failure Modes	Effects of Failure	Causes of Failure	Recommended Action	SR	Ref.
Account System	1. User cannot log in 2. Patient is mistakenly given access to the clinician view	1. The clinician cannot reach patient records or assessments, which delays annotation. 2. Other patients' recordings, annotations, and assessment results are exposed and could be altered, which is a potential violation of HIPAA.	1. Incorrect login credentials, email not yet verified, or authentication service unavailable. 2. Role is not checked or is checked incorrectly, or the wrong role is assigned at sign-up.	1. Show error messages that specify the cause, including when the service is unavailable. 2. Check the account's role on every server request, and deny access by default unless the role permits it.	1. NFR2 2. NFR5, NFR15	1. HA-AS1 2. HA-AS2
Web Platform	1. Annotation progress is lost between sessions 2. Automated annotations cannot be distinguished from clinician-reviewed content	1. Clinicians must spend time redoing annotation work. 2. Clinicians rely on unchecked transcripts or flags in their reports.	1. No autosave or save prompts, session timeout, or browser closed. 2. Missing or unclear labels for unverified content.	1. Autosave on each change and when navigating between tasks, and warn the user about unsaved changes before they leave. 2. Visually label unverified segments, scores, and flags, and show the number of unverified items before the annotation can be marked complete.	1. NFR21, FR31 2. NFR14	1. HA-WP1 2. HA-WP2

Concluded

6 Safety and Security Requirements

[Newly discovered requirements. These should also be added to the SRS. (A rationale design process how and why to fake it.) —SS]

The following requirements were identified through the FMEA in Section 5 and were not originally covered by the existing requirements in the SRS.

6.1 Security Requirements

HA-SER1. The system shall only send the audio track of a recording to the external transcription service, with no patient-identifying information.
Rationale: Since OpenAI may keep submitted data for up to 30 days (**?**), it is crucial to control what data the system sends them. Sending only audio without identifiers limits the harm if OpenAI's copy is ever exposed.
Fit Criterion: Inspection of every request sent to the transcription service must show that it contains no video frames and no patient names or identifiers in the file name or metadata.

HA-SER2. The system shall protect all database queries and displayed user input against injection attacks.
Rationale: Free-text fields, such as notes and transcript edits, accept arbitrary input. This could be used to read or alter highly sensitive patient records through SQL injection.
Fit Criterion: All database access shall be handled using a well-known secure method, such as parameterized queries or an Object-Relational Mapping (ORM) tool. Queries shall not be built by concatenating user input.

HA-SER3. The system shall verify on the server that the logged-in clinician is authorized to access a patient's data before responding to any request for it.
Rationale: While NFR5 controls which patients a clinician sees listed, it does not specify which records the server returns. Changing an ID in a request could expose another clinician's patients, which is a violation of privacy.
Fit Criterion: Every request for a patient's records, reports, or recordings shall be rejected unless that patient is assigned to the logged-in clinician.

6.2 Safety Requirements

HA-SAR1. The system shall allow clinicians to continue reviewing and annotating assessments when the transcription service is unavailable.
Rationale: Transcription relies on an external service outside the team's control, so an outage must not stop clinicians from reviewing recordings or editing saved transcripts.
Fit Criterion: When the transcription service cannot be reached, clinicians shall be able to view recordings, edit previously saved transcripts, add annotations and notes, and transcribe a recording manually. Each failed transcription job will display its status to the clinician and be retried automatically.

HA-SAR2. The system shall prevent an assessment report from being finalized if it contains automated content that the clinician has not verified.

Rationale: Clinicians must make the final judgement on any auto-generated content to ensure a model error does not appear as a finding in a final report. Errors in transcription or report generation can easily result in a misleading or incorrect speech diagnosis, which has potential to cause great harm.
Fit Criterion: The system shall not allow a report to be marked final or downloaded until every automated transcript segment, score, and flag included in it has been accepted or corrected by the clinician. Any report that has not been finalized shall display a visible label stating that it contains unverified content.

7 Roadmap

From conducting the hazard analysis, a set of new safety and security requirements were identified. Due to the time constraints of the capstone project, not all of them will be implemented at this stage. TeLaC will prioritize HA-SER1, HA-SER2, HA-SER3, and HA-SAR2, as they protect sensitive patient data and ensure that unverified model output never appears in a clinical report. HA-SER1 and HA-SER3 is thought to be implemented during the process of integrating SAFA into TeLaC, since this is when recordings are first sent to the transcription service and patient data is first requested through the platform. HA-SAR1 will be addressed in future development phases after the capstone timeline, since transcription service outages are expected to be rare and clinicians can still review manually and previously transcribed assessments in the meantime.

8 Appendix — Generative AI Use Disclosure

8.1 AI Tools Used

- Claude Sonnet 5 (Medium)
- ChatGPT-5.6 Luna

8.2 How and Where Used

The following uses of AI apply throughout the entire document. If a section is not listed below, no additional AI use occurred in that section.

- Fixing grammar and spelling
  - *Prompt: "can you fix the grammar and spelling in this section but keep the wording close to mine: [insert section text]"*
- Performing a final check over the document at the end, comparing each section against the original document comments and the rubric to ensure nothing was missed
  - *Prompt: "here is the rubric for this deliverable: [insert rubric]. here is the document: [insert document code]. can you go through each section and check it against the rubric, and tell me if we missed anything"*
- Organizing what each team member used AI for (listed as bullet points) into the corresponding sections of this appendix
  - *Prompt: "here is a list of how everyone on the team used ai: [insert bullet points]. can you organize these into the sections of the document they belong to"*
- Debugging possible LaTeX errors
- Creating references
  - *Prompt: "can you format this in a citation for me: [insert link]. here is an example: [insert example citation]"*

Section 1: Introduction

- Claude was used to fix the wording and flesh out the introduction.
  - *Prompt: "can you elaborate on this intro and make it sound better: [insert introduction]"*

Section 2: Scope and Purpose of Hazard Analysis

- Claude was used to fix the wording and flesh out the scope and purpose.
  - *Prompt: "can you help me re-word and fix this: [insert paragraph]. the question being asked: [insert question from the template]"*

Section 3: System Boundaries and Components

- Claude was used to fix the wording of the component descriptions.
  - *Prompt: "can you fix the wording: [insert component description]"*

Section 4: Critical Assumptions

- Claude was used to help brainstorm critical assumptions.
  - *Prompt: "what are some critical assumptions we could make for the system: [insert system description]"*
- Claude was used to fix the wording of the assumptions.
  - *Prompt: "can you fix the wording: [insert assumptions]"*

Section 5: Failure Mode and Effect Analysis

- Claude was used to put the group's data into the LaTeX table.
  - *Prompt: "can u make a table with this format: [insert example table] but remove all the data in it and put this instead: [insert data]"*
- Claude was used to help brainstorm failure modes for the components.
  - *Prompt: "can you help me identify failure modes for [insert component]"*

Section 6: Safety and Security Requirements

- Claude was used to help brainstorm safety and security requirements.
  - *Prompt: "can you help me come up with safety and security requirements for the failure modes in this table: [insert FMEA table]"*
- Claude was used to check the wording of the requirements.
  - *Prompt: "can you check the wording of these requirements: [insert requirements]"*

Section 7: Roadmap

- Claude was used to check the wording.
  - *Prompt: "can you check the wording of this: [insert roadmap]"*

8.3 Verification of AI-Generated Content

To verify the output of the LLM, each team member went over their section and presented their work to the group, explaining what they had completed since the previous meeting. Each member was able to successfully explain their sections. Additionally, before any pull request was approved, a team member who had not written the section reviewed the work to ensure it was cohesive, accurate, and consistent with the rest of the document.

The factual content of the documents came from the research of the group members. The writer of each section researched, obtained, and documented all referenced sources. After the models suggested wording improvements, the writer re-read the section to confirm that it still conveyed the same point and that the research was not altered.

9 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Nour

Question 1

What went well with this deliverable was determining possible failure modes regarding the ML Transcription model. There are several things to be wary of when dealing with external APIs and using machine learning models that have been tuned for our case. This is because they are not always predictable. Models typically have some form of evaluation, but the model was already chosen by our supervisors, and seeing that it has not been evaluated was a bit concerning since it is hard to be certain how well such a model will perform.

What reassures me is that speech clinicians can go back and verify the model's output against the video playback, which is extremely important. However, this places the responsibility on the clinician, including how much time they are willing to spend annotating and whether they will watch the entire assessment instead of skimming through it.

Another worrying issue is data retention. If Zero Data Retention was not applied for when the team started using OpenAI's Whisper model, client information has been stored by OpenAI. OpenAI doesn't use this data to train its models unless a user explicitly agrees to share it, but the data is still stored on its servers. OpenAI is a large and popular company that could experience cyberattacks, so having sensitive data stored in multiple places is not ideal.

Question 2

This deliverable was relatively straightforward, and our team did not run into any major issues while completing it. We began by brainstorming the components of the system together, and we

were able to identify what could go wrong with each one and possible hazards that could result. TeLaC is a privacy-critical system, so security and data breaches were extremely important for us to address. The system also depends on performance and quality for different features to work, which made it easy to determine its high-level components. Because we share a strong understanding of the system, prioritizing the safety requirements was not difficult. Our supervisors also mentioned which parts of the system they wanted completed first, which gave us a clear direction as to which parts of the system were the most important.

Catherine

Question 1

As a team, hazards and risks had already been heavily discussed throughout the project, so during meetings, we were able to easily brainstorm and discuss each section of the deliverable, and divvy up all the parts. As we now had many more resources, it was also easier to ask more guided questions when we were confused or not aligned with one another on a concept related to our project, so communication was also a lot clearer for this deliverable. Pre-splitting the system into components also made it easy to divide the work between us, since each person could focus on the parts they were most familiar with.

Question 3

Before starting this deliverable, we had already thought about and discussed in pretty deep detail different risks and hazards to consider, especially for a project that touches real patients and health practices. For instance, as mentioned throughout the deliverable, because the system stores video recordings and health records of children, privacy and data security were something we talked about extensively from the start. They also came up in our previous deliverables. We also knew that the transcription model could have trouble differentiating between speakers, different languages, or sentences in general, so we knew that a licensed clinician would always have to be in the loop to review and verify everything.

Other risks only came up while doing the Hazard Analysis, where going through each component made us think about how each part could fail on its own, instead of just the system as a whole. For example, while writing about the database and report generation module, we realized a clinician could potentially view or edit the wrong patient assessment, or lose important edits without knowing, resulting in incorrect results. Writing the critical assumptions also brought up risks we hadn't considered, like what happens if the internet cuts out partway through an assessment, or if the child happens to need a lot of help from a parent during their assessment and the parent happens to interfere too much. Trying to decide whether these should be assumptions or actual hazards made us think more about what we could realistically control and, if so, how.

Yilin

Question 1

Since our patient-facing platform was originally developed by a previous year's capstone team, reading their hazard analysis was helpful, particularly for judging whether the hazards we were considering fell within the scope of our project. With this strong reference, the team was able to divide the document efficiently, and each member worked independently until we came together for

a document review. Working in two rounds, first individually and then as a group, was effective for iterating on and improving our initial work.

Javayria

Question 1

Overall, our team was much better prepared for this deliverable. It was directly relevant to our project given that we will be handling protected health records and audio recordings of children. Because we had already spent a large amount of time brainstorming and discussing security concerns with our supervisors while drafting the SRS, the Hazard Analysis flowed naturally. Since we are becoming more knowledgeable and aware of associated risks, we were able to easily identify potential failure modes and define additional safety and security requirements. Our supervisors also set clear expectations for security by saying that data protection should be our highest priority. While other safety or security features can be added after the core integration and functionality are complete, enforcing authorized access and defending against attacks are absolutely non-negotiable. Having priorities early on is extremely helpful, and will give us enough time to research and implement them properly.

Question 4

Two other types of risk in software products are ethical risks and privacy/security risks. Although they are often not given the highest priority, ethical risks are important to consider because software is never isolated. It is built to be used by real people, and the decisions it makes or supports affect their lives in many ways. Detection software is especially prone to this type of risk, since it often relies on pre-trained models that may be biased by the data they were trained on. A model that performs worse for certain population groups can treat those users unfairly without anyone noticing. Privacy and security risks are also very important to consider because software often handles large amounts of user data that must be protected from being leaked or accessed by unauthorized people. Information such as Social Insurance Numbers, home addresses, and a person's health or medical history is confidential, and exposing it can lead to identity theft. An example where this is extremely important to consider is banking software. A privacy breach could lead to financial theft, where an attacker gains access to someone else's assets and financial information. This can cause the victim of such an attack to lose money or be locked out of their own accounts.